

ECE 332 Fall 2025

Exam 1

(100 pts total)

November 6, 2025

Allowed: Calculator approved for the FE exam and a 2-sided page of notes.

Not allowed: Any other devices or materials.

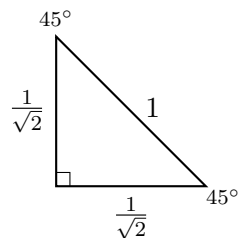
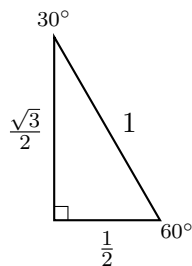
Use the approximations

$$\epsilon_0 \approx \frac{1}{36\pi} \times 10^{-9} \text{ F/m}, \quad \mu_0 = 4\pi \times 10^{-7} \text{ H/m}, \quad \eta_0 \approx 120\pi \ \Omega$$

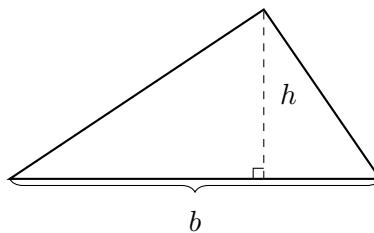
in your calculations.

Useful Facts:

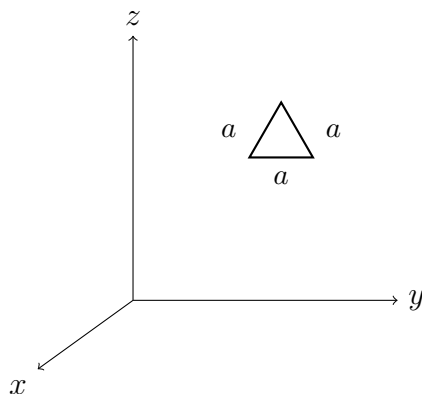
“Special” right triangles:



The area of a triangle is $A = \frac{1}{2}bh$, where b is the length of the “base” and h is the “height.”



1. (20 points total) Suppose an equilateral triangular conducting loop of N turns with sides of length a is located in the y - z plane with normal $d\mathbf{s} = \hat{x} ds$, as shown below,



in a magnetic field

$$\mathbf{B}(t) = B_0(-2\hat{x} + 3\hat{y}) \sin(\omega t).$$

(a) Find a formula for the magnetic flux through one turn of the loop. (6 points)

(b) Find a formula for the emf induced in the loop at time t . (6 points)

(c) Determine the current flowing through a $1\text{ k}\Omega$ resistor incorporated in the loop when $N = 10$, $a = 8\text{ cm}$, $B_0 = 0.02\text{ T}$, $\omega = 2\text{ MHz}$, and $\omega t = \pi/4$, assuming that the resistance of the loop material is negligible. (8 points)

2. A straight conducting wire of length $a = 15$ cm lying along the $\hat{y}$ direction moves with velocity $\mathbf{u} = 12\hat{x}$ cm/s in a magnetic field

$$\mathbf{B} = (0.1\hat{x} - 0.05\hat{z}) \text{ T}.$$

Find the emf between the two ends of the wire. (12 points)

3. Suppose a total conduction current $I_c = 2 \sin(\omega t)$ μA flows through a column of distilled water with cross-sectional area A in a glass tube. Distilled water has conductivity $\sigma \approx 10^{-4}$ S/m and relative permittivity $\epsilon_r \approx 26\pi$ and the current flowing through the glass is negligible compared to the current flowing through the water. (12 points total)

- (a) Use the relationship between the conduction current, the current density, the cross-sectional area, and Ohm's law to determine the electric field intensity E at a cross-section of the water. (6 points)

- (b) Determine the displacement current I_d through the cross-section when the angular frequency of I_c is $\omega = 10^9 \text{ rad s}^{-1}$ and $\omega t = \pi/3$. (6 points)

4. Distilled water has a conductivity $\sigma \approx 10^{-4} \text{ S/m}$ and $\epsilon_r \approx 26\pi$. (10 points total)

(a) Find the relaxation time constant for distilled water. (5 points)

(b) How long does it take for an excess charge density in distilled water to decay to e^{-3} times its initial value? (5 points)

5. (30 points total) Suppose the electric field intensity of an EM wave in a nonconducting, source-free medium with $\epsilon_r = 6$ and $\mu_r = 1$ is

$$\mathbf{E}(z, t) = 4 \cos\left((6\pi \times 10^8 \text{ rad/s})t - kz - \frac{\pi}{6}\right) \hat{y} \text{ V/m}.$$

- (a) Determine the angular frequency ω , the wavenumber k , the wavelength λ , and the intrinsic impedance η . (10 points)

- (b) Find $\tilde{\mathbf{E}}(z)$. (5 points)

- (c) Find $\tilde{\mathbf{H}}(z)$. (5 points)

(d) Determine the magnetic field intensity $\mathbf{H}(z, t)$. (5 points)

(e) Determine the average power density that the EM wave carries. (5 points)

6. Consider a $\frac{2}{\pi}$ GHz EM wave traveling in distilled water, where $\sigma \approx 10^{-4}$ S/m, $\epsilon_r \approx 26\pi$, and $\mu_r = 1$. The wave travels downward into the water, where the $+z$ direction is down and $z = 0$ is just below the surface. The amplitude of the electric field at the surface is 4 V/m. (16 points total)

(a) Find ϵ''/ϵ' . (6 points)

(b) What type of medium is distilled water at this frequency? Justify your answer. (5 points)

(c) Find the skin depth for distilled water at this frequency. (5 points)